

|    |   |                                                                                                                                                                                                                                    |
|----|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |   | $\left( \frac{r}{2} \right) r$                                                                                                                                                                                                     |
| 10 | D | <p>Gravitational force provides the centripetal force for the satellite to orbit the Earth</p> $\rightarrow \frac{GMm}{r^2} = ma \rightarrow a = \frac{GM}{r^2}$ $\frac{GMm}{r^2} = ma = \frac{mv^2}{r}$ $v = \sqrt{\frac{GM}{r}}$ |
| 11 | A | $\sqrt{\frac{GM}{r}}$                                                                                                                                                                                                              |